LeetQuiz 3 - Refined Questions

24 unique questions. Duplicates removed across all four PDFs. Based on the May 2026 exam guide.

Updated product and SQL object names have a yellow background. Substantive corrections are disclosed below each affected question. Original page references refer to archived screenshot PDFs.

Question 40 - Transformation

Original: leetquiz-3.pdf, page 2

A pipeline uses Python for Bronze/Silver and SQL for Gold, with streaming input. When moving to Lakeflow Spark Declarative Pipelines, which architecture change is required?

A. No change to its language mix, medallion architecture, or streaming source is required.

B. It must stop using the medallion architecture.

C. It must be rewritten entirely in SQL.

D. It must replace streaming input with a batch source.

E. It must be rewritten entirely in Python.

Source correction: Clarified that no listed architecture change is required, not that the code needs no migration work.

Question 41 - Transformation

Original: leetquiz-3.pdf, page 3

Which join includes every student, even those with no course enrollment?

```
SELECT students.name,
students.age, enrollments.course_id
FROM students
_____ enrollments
ON students.student_id = enrollments.student_id
```

A. RIGHT JOIN

B. INNER JOIN

C. LEFT JOIN

D. ANTI JOIN

E. CROSS JOIN

Question 42 - Transformation

Original: leetquiz-3.pdf, page 4

A data entity combines data from two tables, must be reusable in other sessions, and must store results physically. Which object should be created?

A. Table

B. Function

C. View

D. Temporary view

Question 43 - Transformation

Original: leetquiz-3.pdf, page 11

A DataFrame df has nested field address.city. Which code adds a top-level city column while retaining the other existing columns?

A. `df.withColumn('city', col('address.city'))`

B. `df.select(col('address.city').alias('city'))`

C. `df.withColumn('city', 'address.city')`

D. `df.select('address.city').alias('city')`

Source correction: Specified retaining existing columns: the original select-and-alias choice also extracts a top-level field and was otherwise valid.

Question 44 - Ingestion

Original: leetquiz-3.pdf, page 14

Which workload model does Auto Loader use to incrementally process files?

A. Structured Streaming workloads

B. Machine learning training as its ingestion API

C. Every serverless workload regardless of API

D. Only conventional spark.read batch workloads

E. Dashboard rendering workloads

Source correction: Avoided the original misleading batch-versus-streaming dichotomy: Auto Loader can support batch-style scheduled processing using AvailableNow.

Question 45 - Ingestion

Original: leetquiz-3.pdf, page 15

A streaming ingestion job must process all currently available input in one micro-batch and then stop. Fill the blank:

```
(spark.readStream.table("sales")
.withColumn("avg_price", col("sales") / col("units"))
.writeStream
.option("checkpointLocation", checkpointPath)
.outputMode("append")
._____
.toTable("new_sales"))
```

A. `trigger(once=True)`

B. `trigger(continuous="once")`

C. `processingTime("once")`

D. `trigger(processingTime="once")`

E. `processingTime(1)`

Source correction: Corrected source readStream, non-aggregated append output mode, and toTable sink. once=True is legacy/deprecated; retained its ingestion concept with an explicit explanation.

Question 46 - Platform

Original: leetquiz-3.pdf, page 16

Which describes the storage organization of a Delta table?

A. One file containing all data, history, and metadata.

B. One data file with metadata in an unrelated location.

C. A collection of data files and a transaction log that records table metadata and versions.

D. A collection of files containing only data, without metadata.

E. One file containing only table data.

Question 47 - Lakeflow Jobs

Original: leetquiz-3.pdf, page 19

In a multi-task Lakeflow Job, how can Task A share a generated ID with downstream Task B?

A. `dbutils.jobs.taskValues.set(key="generated_id", value=generated_id)`

B. Write a DBFS file as the only supported method.

C. Set cluster environment variables for task-to-task values.

D. Notebook widgets automatically pass values between every task.

Question 48 - Transformation

Original: leetquiz-3.pdf, page 26

A Delta table sales is used by SQL analysts. Which command reads this registered table as a PySpark DataFrame?

A. `SELECT * FROM sales`

B. There is no way to share data between PySpark and SQL.

C. `spark.sql("sales")`

D. `spark.delta.table("sales")`

E. `spark.table("sales")`

Question 49 - Transformation

Original: leetquiz-3.pdf, page 28

A dataset has transaction_date strings in yyyy-MM-dd format. Which query casts them to timestamps and extracts year, month, and day from the timestamp?

A. `SELECT CAST(transaction_date AS TIMESTAMP), EXTRACT(YEAR FROM transaction_date), EXTRACT(MONTH FROM transaction_date), EXTRACT(DAY FROM transaction_date) FROM dataset`

B. `SELECT transaction_date, YEAR(transaction_date), MONTH(transaction_date), DAY(transaction_date) FROM dataset`

C. `SELECT CAST(transaction_date AS TIMESTAMP) AS ts, EXTRACT(YEAR FROM ts) AS year, EXTRACT(MONTH FROM ts) AS month, EXTRACT(DAY FROM ts) AS day FROM dataset`

D. `SELECT FROM_UNIXTIME(CAST(transaction_date AS TIMESTAMP), 'yyyy-MM-dd') AS formatted_date, EXTRACT(YEAR FROM formatted_date) AS year FROM dataset`

Question 50 - Governance

Original: leetquiz-3.pdf, page 30

Which grouping best identifies Unity Catalog data hierarchy objects plus the securable used to govern a cloud storage path?

A. Catalog, schema, table, and external location

B. Catalog, schema, table, and data format

C. Catalog, schema, table, and data type

D. Catalog, schema, table, and a generic source label

Source correction: Replaced the original Database/Data Location vocabulary with the actual Unity Catalog objects, not a cosmetic product rename.

Question 51 - Governance

Original: leetquiz-3.pdf, page 31

Assuming required USE CATALOG and USE SCHEMA privileges already exist, which command grants a principal read access to the orders table?

A. `GRANT USAGE ON TABLE main.sales.orders TO `user@example.com`;`

B. `GRANT MODIFY ON TABLE main.sales.orders TO `user@example.com`;`

C. `GRANT SELECT ON TABLE main.sales.orders TO `user@example.com`;`

D. `GRANT READ ON TABLE main.sales.orders TO `user@example.com`;`

Source correction: Corrected principal quoting and stated parent-usage prerequisites.

Question 52 - Transformation

Original: leetquiz-3.pdf, page 33

customers contains registered customers and orders contains order_customer_id. Which query finds customers who have placed no orders?

A. `SELECT customer_id FROM customers INTERSECT SELECT order_customer_id FROM orders`

B. `SELECT customer_id FROM customers UNION SELECT order_customer_id FROM orders`

C. `SELECT c.customer_id FROM customers c INNER JOIN orders o ON c.customer_id = o.order_customer_id`

D. `SELECT customer_id FROM customers UNION ALL SELECT order_customer_id FROM orders`

E. `SELECT customer_id FROM customers EXCEPT SELECT order_customer_id FROM orders`

Question 53 - Ingestion

Original: leetquiz-3.pdf, page 34

Which is NOT a feature of Auto Loader?

A. Detect schema changes and rescue unexpected data.

B. Provide a SQL warehouse for running general Databricks SQL queries.

C. Incrementally process new files arriving in cloud storage.

D. Use file notifications to avoid repeated directory listing.

E. Track discovered files in checkpoint state for exactly-once ingestion to Delta.

Question 54 - Lakeflow Jobs

Original: leetquiz-3.pdf, page 35

Three source notebooks define dependent datasets. How should the engineer set up a Lakeflow Spark Declarative Pipelines pipeline to manage them?

A. Create the pipeline only from the Data page.

B. Create a pipeline from Jobs & Pipelines and add the source code files.

C. Create the pipeline only from the Compute page.

D. Rewrite every source in Python; SQL is unsupported.

E. Rewrite every source in SQL; Python is unsupported.

Source correction: Updated creation UI and clarified declarative dataset dependencies instead of notebook execution ordering.

Question 55 - Transformation

Original: leetquiz-3.pdf, page 36

Which command creates an empty Delta table with the required schema even if a table of that name already exists? Existing contents may be replaced.

A. `CREATE OR REPLACE TABLE table_name (employeeId STRING, startDate DATE, avgRating FLOAT) USING DELTA;`

B. `CREATE OR REPLACE TABLE table_name WITH COLUMNS (employeeId STRING, startDate DATE, avgRating FLOAT) USING DELTA;`

C. `CREATE TABLE IF NOT EXISTS table_name (employeeId STRING, startDate DATE, avgRating FLOAT) USING DELTA;`

D. `CREATE TABLE table_name AS SELECT employeeId STRING, startDate DATE, avgRating FLOAT;`

Source correction: Merged equivalent empty-table variants and clarified that existing contents may be replaced.

Question 56 - Transformation

Original: leetquiz-3.pdf, page 37

Why is STREAM used in this Lakeflow Spark Declarative Pipelines query?

```
CREATE OR REFRESH STREAMING TABLE loyal_customers AS
SELECT customer_id FROM STREAM(customers)
WHERE loyalty_level = 'high';
```

A. STREAM is unnecessary and always causes an error.

B. It means the target is only a batch materialized view.

C. It reads the source using streaming semantics so new source rows are processed incrementally.

D. It means the source must be a Python DataFrame variable.

E. It guarantees the source changed since the last run.

Source correction: Replaced legacy LIVE syntax and corrected the original claim that STREAM proves the source itself is a streaming live table.

Question 57 - Optimization

Original: leetquiz-3.pdf, page 39

Delta data files are very small. Which keyword compacts them into larger files?

A. OPTIMIZE

B. VACUUM

C. COMPACTION

D. REPARTITION

Question 58 - Optimization

Original: leetquiz-3.pdf, page 41

Which valid command removes unreferenced files from sales that are older than the seven-day retention threshold?

A. `VACUUM sales RETAIN 7 DAYS`

B. `VACUUM sales RETAIN 604800 SECONDS`

C. `VACUUM sales RETAIN 168 HOURS`

D. `VACUUM sales RETAIN 10080 MINUTES`

E. `VACUUM sales RETAIN 1 WEEK`

Source correction: Clarified unreferenced-file retention rather than claiming all files older than one week are deleted.

Question 59 - Platform

Original: leetquiz-3.pdf, page 42

In which scenario could a team use classic compute instance pools?

A. An automated report needs fast compute startup to refresh quickly.

B. An automated report needs to be reproducible.

C. An automated report needs tests for logic errors.

D. An automated report needs version control.

E. An automated report needs stakeholder access control.

Question 60 - Governance

Original: leetquiz-3.pdf, page 44

Mike owns a Unity Catalog sales table. Ross owns a view over sales, and David has SELECT on the view plus its parent usage privileges. They query on a SQL warehouse. Ross later loses SELECT on sales. What restores view access?

A. Mike must transfer table ownership to Ross.

B. David must become the view owner.

C. Restore Ross, the view owner, access to sales and its required parent namespaces.

D. David must own sales.

E. David always needs direct SELECT on sales on a SQL warehouse.

Source correction: Corrected the old consumer-needs-base-table-SELECT premise for Unity Catalog SQL warehouses and stated the actual failing permission.

Question 61 - Optimization

Original: leetquiz-3.pdf, page 45

VACUUM removed some unused files but left others. What retention-related explanation is most likely?

A. Deleted files were larger than a size threshold.

B. Deleted files were newer than the retention threshold.

C. Deleted files were older than the retention threshold; remaining unused files were too recent.

D. Deleted files were smaller than a size threshold.

E. VACUUM never uses retention rules.

Question 62 - Transformation

Original: leetquiz-3.pdf, page 49

A relational object must be reusable across sessions without physically storing the query results. Which object should be created?

A. View

B. Temporary view

C. Delta table

D. Database

E. Common table expression scoped to one query

Question 63 - Optimization

Original: leetquiz-3.pdf, page 50

The orders Delta table is partitioned by order_date, stored as DATE. Which command compacts only its 2022 partitions?

A. `OPTIMIZE TABLE orders WHEN order_date >= DATE '2022-01-01'`

B. `OPTIMIZE orders WHERE order_date >= DATE '2022-01-01' AND order_date < DATE '2023-01-01'`

C. `OPTIMIZE TABLE orders WHERE order_date >= DATE '2022-01-01'`

D. `OPTIMIZE orders FILTER BY order_date >= DATE '2022-01-01'`

E. `OPTIMIZE TABLE orders USING order_date >= DATE '2022-01-01'`

Source correction: Added partition prerequisite and an upper date bound so the query really targets 2022 only.